Synergizing Acoustic and Wi-Fi Signals for Device-Free Gesture Recognition

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***Abstract*—Gesture recognition has significant applications in areas such as assisted living, e-health, and human-device interactions. Moving beyond conventional computer vision techniques, recent studies have increasingly adopted ubiquitous methods like Wi-Fi and acoustic signals, which provide a cost-effective solution for device deployment. In this paper, we explore these two ubiquitous techniques to enhance gesture recognition, focusing on overcoming challenges associated with multi-modal fusion. To harmonize information from these inherently different signal types, we propose a tailored fusion strategy specifically designed for Wi-Fi and acoustic signals. Traditional multi-modal fusion methods often lack a theoretical framework due to insufficient analysis of the fundamental characteristics of different signals. To address this gap, we introduce the concept of the Hybrid Zone, a novel theoretical framework that models the interaction and fusion of acoustic and Wi-Fi sensing signals. The Hybrid Zone offers a unified perspective on the interaction between acoustic and Wi-Fi sensing areas and delivers insights into the granular synthesis of their velocity profiles. Our experimental results demonstrate strong performance, achieving a gesture recognition accuracy rate of 94.69%.**

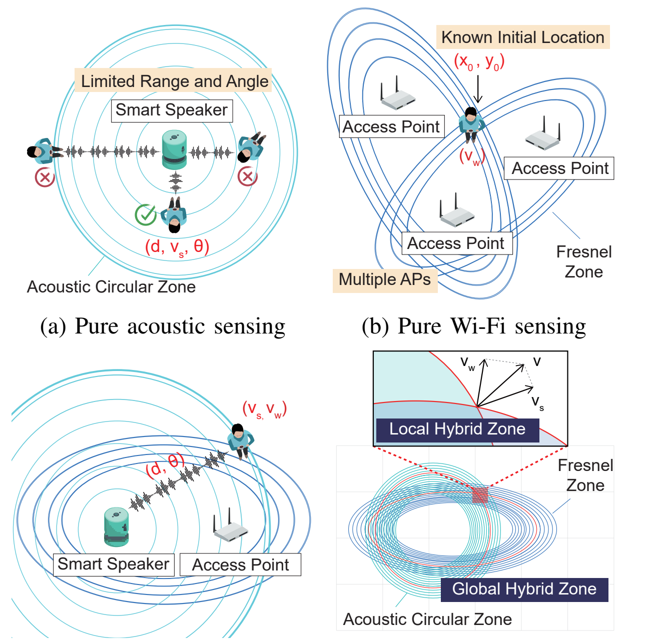
***Index Terms*—Acoustic, Wi-Fi, Gesture recognition**

I. Introduction

A. Background and Motivation

The gesture recognition market, without requiring physical touch, is projected to reach 37.6 billion by 2026 [1]. The growth in non-contact sensing technology has enabled smart devices to detect human actions without direct contact, enhancing human-computer interaction by providing more natural and intuitive control methods. In this context, Wi-Fi [2]–[6] and acoustic signal-based sensing [7]–[9], both integral components of household wireless technologies, are receiving increased focus. Their ubiquity in homes, serving as the backbone for internet and audio systems, renders them highly suitable for everyday applications in non-contact human action detection. Despite their potential, standalone sensing methods have limitations: i) Acoustic systems [10] are advantageous in low-light or obstructed environments; however, they are susceptible to ambient noise and echoes [7], [8], and their accuracy decreases with distance. These systems also necessitate a direct line-of-sight to the speaker. ii) Wi-Fi-based systems offer longer sensing ranges compared to acoustic systems. Nevertheless, they require specific knowledge of the user's location for accurate feature extraction, and a minimum of three Wi-Fi access points are needed for location-independent recognition [2], [11], [12].

These challenges in pure technology-based gesture recognition led us to investigate the fusion of acoustic and Wi-Fi sensing technologies. This approach can be practically implemented using common household devices such as smart speakers and Wi-Fi routers, which are typically interconnected. This fusion aims to overcome the individual limitations of each technology, offering a more robust and accurate gesture recognition system. Naturally, we explore the fusion of acoustic and Wi-Fi sensing technologies in Fig. 1(c). It is practical to build the described fusion sensing scenarios using just one pair of devices, such as the smart speaker and the router, because voice devices commonly found in our homes can usually connect to the Wi-Fi router.



(a) Pure acoustic sensing (b) Pure Wi-Fi sensing

(c) Acoustic & Wi-Fi sensing (d) Hybrid Zone

Fig. 1: Rapid movement affects distance assessment accuracy.

Currently, many cross-modal fusion approaches rely heavily on the cross-attention mechanism [13]–[15], which can yield excellent results. The cross-attention mechanism allows one set of features to attend to another set by computing attention scores between them, enabling effective information exchange between different modalities. However, they suffer from three main issues: i) the need for manual collection of training datasets, ii) lack of explanation for the underlying mechanisms, and iii) difficulty in ensuring the generalization capability of the system. Some recent works have explored the fusion of Wi-Fi and acoustics in the model [16], but they require targets or users to actively emit sounds (*e.g.*, footsteps). However, when making gestures, people do not produce sounds, so acoustic devices are needed to actively emit inaudible signals to perceive the surrounding targets.

In contrast to sound source localization methods [16], our contact-free sensing approach focuses on analyzing signals reflected from targets. Nevertheless, compared to direct signals, reflected signals have weaker strength, presenting a greater challenge in achieving this objective.

B. Challenges and Solutions

This paper presents *Hybrid Zone*, a model to illustrate how to fuse acoustic and Wi-Fi sensing techniques. The basic principle consists of global and local stages: i) The global Hybrid Zone defines the overall distribution of the hybrid sensing signal. We leverage the acoustic signal to approximate the target's position, enabling us to focus on the specific area of interest, referred to as the local Hybrid Zone; ii) At the local stage, the Hybrid Zone accurately synthesizes user velocity. This is achieved by combining the velocity measurements from both acoustic and Wi-Fi signals, leading to enhanced precision in tracking and recognition. The study addresses two main challenges in implementing this model.

Challenge 1: How to obtain accurate acoustic features in non-ideal situations? Specifically, non-ideal situations include long distances, fast motion, and indirect orientation. In the case of long-distance and non-direct directions, the resulting signal is weaker, making it harder to accurately extract features. Additionally, measuring the distance of a fast-moving target can be challenging due to the Doppler effect, which causes an extra frequency shift in the reflected signal that hinders the measurement of the distance. To address these problems, we first utilize the triangular chirp signal to distinguish the Doppler effect and the target's distance. Next, we design a signal enhancement method so that the weak reflected signal can be enhanced for accurate feature extraction.

Challenge 2: How to design a model to fuse multi-modal information for fine-grained behavior tracking? Previous research has typically employed neural networks, such as the cross-attention mechanism [14], to combine sensing features from different modalities. However, there is a gap in the theoretical understanding of how these modalities are integrated. In order to provide theoretical guidance for the fusion of Wi-Fi and acoustic sensing technology, we propose the concept of Hybrid Zone. The global Hybrid Zone has revealed that the sensing zone is composed of a cluster of ellipses and a cluster of circles. The altered features with respect to the object are caused by its intersection with this zone. The local Hybrid Zone reveals that the detected velocity features are the components of the actual velocity on the circle and ellipse normal vectors, thus providing a theoretical velocity synthesis model.

C. Contributions and Advantages

In this paper, we make the following contributions:

- To the best of our knowledge, Hybrid Zone is the first to provide a theoretical explanation for the fusion of acoustic and Wi-Fi sensing zones. Through the design of both global and local Hybrid Zones, we can fully leverage the benefits of acoustic and Wi-Fi sensing to achieve precise gesture recognition in a complementary way.
- To obtain stable acoustic features, we propose a series of methods at the signal processing level, including triangular chirp analysis and signal enhancement, enabling accurate position-related features to be resolved even in non-ideal situations.
- We implement the prototype with commercial off-the-shelf (COTS) Wi-Fi and acoustic devices. The experimental results demonstrate that we can achieve a 93.75% accuracy rate in gesture recognition.

With the proposed technique contribution, our system offers a distinct advantage over previous systems: voice-controlled devices typically have the ability to connect to the Wi-Fi router, which allows for the complementary use of Wi-Fi and acoustics without extra constraints. Furthermore, constructing a theoretical model can enhance our understanding of the sensing mechanism and provide theoretical guidance for the practical implementation of the system.

II. Related Works

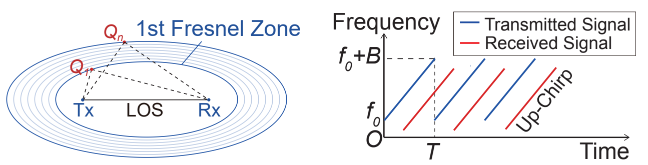
In this study, our methodology avoids many of the pitfalls typically encountered in single-data-source systems. As displayed in TABLE I, we have selected specific criteria for assessing the efficacy of these methods. These criteria encompass their capability to identify swift motions, offer theoretical analysis, and extract location-independent attributes.

Wi-Fi Sensing System. Wi-Fi-based gesture recognition systems operate by identifying human gesture-induced alterations in Wi-Fi signal paths. Each gesture generates distinctive CSI variations, which are learned and recognized by machine learning algorithms. However, their precision can be influenced by device placement, movements of other objects, and shifts in the environment. In particular, Widar3.0 [12] establishes a theoretical model and proposes a body velocity profile (BVP) for cross-domain human gesture recognition. However, it requires multiple Wi-Fi devices (*e.g.*, 6 APs) and a known user's location to extract location-independent features. OneFi [17] designs a lightweight few-shot learning framework to recognize unseen gestures with a shorter training time. WiGr [18] uses a domain-transferable mapping mechanism and a dual-path prototypical network to extract domain-independent features for cross-domain gesture recognition. However, the reliance on neural networks [2], [17]–[20] in these methods is excessive and they lack an analysis of the internal mechanism of sensing ability, making it challenging to determine their scalability.

Acoustic Sensing System. Acoustic-based gesture recognition systems leverage sound waves, frequently in the ultrasonic spectrum, to discern gestures. These systems employ a speaker to emit sound waves, with the microphone array detecting any gesture-induced alterations. Subsequently, machine learning algorithms correlate these wave patterns to particular gestures. Notably, these systems operate independently of visual input, proving beneficial in low-light or visually obstructed scenarios, though their efficiency can be influenced by environmental noise or echo. MilliSonic [21] enhances acoustic-based track-

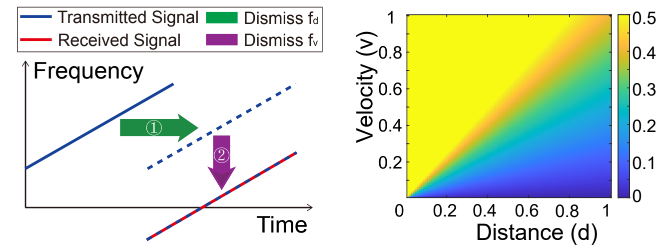
TABLE I: Comparison of different gesture recognition systems.

System	Data Source	Room level?	Fast Move?	Theoretical Model?	Location-Independent?	Single Link?
Widar3.0 2021 [12]	Wi-Fi	Yes	Yes	Yes	Yes	No
OneFi 2021 [17]	Wi-Fi	Yes	No	No	Yes	Yes
WiGr 2022 [18]	Wi-Fi	Yes	No	No	Yes	Yes
FM-Track 2022 [7]	Acoustic	No	No	Yes	Yes	Yes
LASense 2022 [8]	Acoustic	Yes	No	Yes	Yes	Yes
RTrack 2022 [9]	Acoustic	Yes	No	Yes	Yes	Yes
Our Approach	Wi-Fi + Acoustic	Yes	Yes	Yes	Yes	Yes



(a) Schematic diagram of the Fresnel zone. (b) Linear up-chirp transmitted and received signals.

Fig. 2: Theoretical models for Wi-Fi and acoustic sensing.



(a) Shifting for dismissing the distance (f_d) and velocity (f_v). (b) Proportion of velocity's influence on shift relative to distance.

Fig. 3: Rapid movement affects distance assessment accuracy.

ing with a unique algorithm that ensures sub-millimeter precision. Mao *et al.* [10] proposed a joint estimation of the signal's propagation distance and Angle of Arrival (AoA), derived from the 2D MUSIC algorithm. Symphony [22] uses microphone geometry and signal path characteristics to efficiently identify and distinguish signal sources. MAVL [23] is a human voice localization system, which first estimates the room structure and then improves a 3D MUSIC algorithm. Li *et al.* [9] expanded the range of acoustic sensing, demonstrating the possibility of room-scale hand gesture recognition using standard smart speakers. While acoustic methods effectively recognize certain movements, particularly in the tangential direction to the sensor, they struggle with complex, rapid, or multi-dimensional movements. This limitation underscores the need for integrating other sensing technologies for more robust and accurate tracking of fast, remote movements.

III. Preliminaries and Insights

A. Fundamentals

The purpose of this paper is to propose a theoretical model that integrates the use of both Wi-Fi and acoustic sensing. To this end, we will separately describe the theoretical models of these two techniques.

1) The theoretical model for Wi-Fi sensing: The Fresnel zone model [24]–[26] is a fundamental framework in Wi-Fi sensing, explaining how human activity alters Wi-Fi signals. Specifically, human behavior modifies the length of the signal path reflected off the body, expressed as $|T_xQ_n| + |Q_nR_x|$ in Fig. 2(a). A periodic variation in Channel State Information (CSI) magnitude occurs as the path length changes by one wavelength, λ. The Fresnel zone indicates that constant reflection paths form ellipses satisfying

$$|T_xQ_n| + |Q_nR_x| - |T_xR_x| = n\frac{\lambda}{2},$$

where Q_n lies on the n-th ellipsoid. Adjacent ellipses differ in path length by half a wavelength, corresponding to a half-period change in CSI magnitude.

2) The theoretical model for acoustic sensing: The fundamental principle of acoustic sensing is to actively emit inaudible, high-precision Frequency-Modulated Continuous Wave (FMCW) signals and then analyze how the behavior of a person changes the features of the FMCW signal [7]. Generally, these systems [8] typically use chirp signals with a frequency that changes linearly over time, as illustrated in Fig. 2(b). When the chirp signals are reflected by the object, the time delay between the transmitted and received signals can be measured through the frequency offset (red line). Since we can know the frequency slope and sound speed in advance, this time delay can be used to determine the corresponding distance of the object.

Theoretically, if we send the upchirp whose frequency initiates at f_0 and linearly ascends to $f_0{+}B/T$, the transmitted signal, $S_T(t)$, can be mathematically represented by:

$$S_T(t) = \cos\left(2\pi(f_0t + \frac{B}{2T}t^2)\right), \tag{1}$$

where f_0 is the base frequency, B indicates the bandwidth, and T is the period of the chirp. Subsequently, the received signal, $S_R(t)$, maintains the same phase but with a variation in amplitude, α, and a time shift, τ:

$$S_R(t) = A\cos\left(2\pi\left(f_0(t-\tau) + \frac{B}{2T}(t-\tau)^2\right)\right). \tag{2}$$

When we apply the product-to-sum identity to the $S_T(t){\cdot}S_R(t)$ (*i.e.*, $\cos A\cos B = 0.5(\cos(A{-}B){+}\cos(A{+}B))$), and utilize a low-pass filter to filter out the high-frequency signal, we have:

$$S_L(t) = \frac{1}{2}A\cos\left(2\pi\frac{B\tau}{T}t + 2\pi\left(f_0\tau - \frac{B\tau^2}{2T}\right)\right). \tag{3}$$

The frequency of the resulting signal $S_L(t)$ is $B\tau/T$, where τ is the propagation time of the acoustic signal. Finally, we

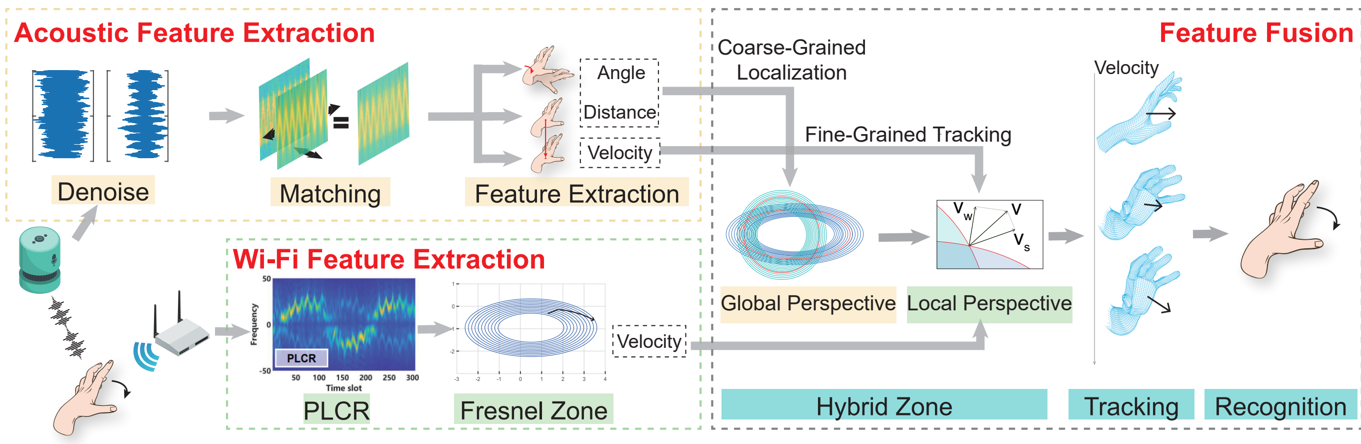


Fig. 4: System overview.

can calculate the corresponding propagation distance through $d = \tau \cdot c_s/2$, where c_s is the sound speed.

B. Observations and Insights

Observation 1: Rapid movement of the target can affect the accuracy of distance assessment.

The accuracy of distance assessment is influenced by the rapid movement of the target due to the combined effects of the Doppler effect and the target's distance. Specifically, the target's distance introduces a horizontal time delay τ in the signal, resulting in a frequency shift:

$$f_d = \frac{B\tau}{T},$$

where B is the bandwidth of the signal and T is the chirp period.

Additionally, the Doppler effect induces a vertical frequency shift given by:

$$f_v = \frac{2fv_s}{c_s},$$

where v_s is the target's velocity, f is the base frequency, and c_s is the speed of sound.

As a result, the total frequency shift combines these two components and is expressed as $f_d + f_v$, rather than just f_d, as previously described in Eq. (3). This adjustment is depicted in Fig. 5(a), which shows how both distance and velocity contribute to the observed frequency changes.

To understand the impact of f_v on f_d, we analyze their ratio, denoted as f_v/f_d, under various scenarios. Fig. 3(b) presents this ratio, where the X-axis represents the target distance, the Y-axis represents the target velocity, and the values indicate f_v/f_d. The results demonstrate that as the velocity-to-distance ratio (v_s/d) increases, the influence of f_v on f_d becomes significant, highlighting the importance of considering f_v in high-velocity or short-distance scenarios.

Mathematically, the ratio f_v/f_d is given by:

$$\frac{f_v}{f_d} = \frac{\frac{2fv_s}{c_s}}{\frac{B\tau}{T}} = \frac{2fv_sT}{Bc_s\tau} = \frac{fT}{B} \cdot \frac{v_s}{d} = k_{coe} \cdot \frac{v_s}{d}. \tag{4}$$

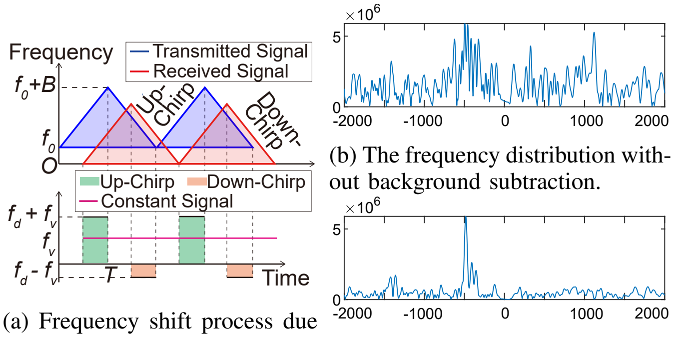
This equation confirms that v_s/d is directly proportional to f_v/f_d, where $k_{coe} = fT/B$. The interdependence of f_d and f_v makes it challenging to extract them separately from a single measurement, necessitating careful consideration in practical implementations.

Insight 1. A straightforward approach to resolving this challenge involves exploring the integral relationship between distance and velocity across different time intervals. However, current methods [7] still encounter the issue of ambiguity, where a unique solution cannot always be obtained. To fundamentally address this limitation, we draw inspiration from the scaling factor in Eq. (4). Since the speed and distance of the person do not change significantly over a short period (*e.g.*, less than 0.1 s), the ratio v_s/d remains relatively constant. Thus, adjusting the scale factor $k_{coe} = fT/B$ effectively modifies the ratio of f_v to f_d, providing a means to control their relative contributions. By designing chirps with respect to different values of f, T, and B, we can systematically adjust how f_d and f_v are mixed in varying proportions. This strategy enables the accurate separation of these two components, resolving the ambiguity and improving the precision of their extraction.

Observation 2: Applicable scenarios of pure acoustic or Wi-Fi-based sensing methods are limited.

i) Acoustic sensing is highly constrained by the user's orientation relative to the smart speaker, requiring the user to face the device at a narrow angle. This significantly limits the range of applicable scenarios. The Doppler velocity in acoustic sensing relies on the projection of the target's true velocity onto the line connecting the target and the acoustic device. When the target is not directly facing the acoustic device, this projection value diminishes. In extreme cases, such as when the target moves tangentially to the circle centered on the device, the Doppler velocity may become undetectable.

ii) Wi-Fi sensing requires accurate prior knowledge of the target's location to extract features that are independent of spatial position. This is because the observed signal features can vary significantly depending on the target's location, even for the same behavior. The underlying mechanism of this variability lies in the distribution of Fresnel zones. As shown



(a) Frequency shift process due to distance and velocity effects during up and down-chirps.

(b) The frequency distribution without background subtraction.

(c) The frequency distribution With background subtraction.

Fig. 5: Background denoise process.

in Fig. 2(a), a denser Fresnel zone leads to more pronounced signal changes, while sparser zones result in weaker variations. This spatial dependency limits the generalizability of Wi-Fi-based sensing in scenarios with unknown or variable target positions.

Insight 2. By leveraging the complementary attributes of Wi-Fi and acoustic signals, we introduce the concept of the ”Hybrid Zone,” which integrates a circular acoustic signal model with an elliptical Wi-Fi-based Fresnel zone model. This framework operates in two distinct stages: i) In the *global Hybrid Zone*, the acoustic signal determines the target's absolute position, enabling precise localization and narrowing the focus to a specific Fresnel ellipse and acoustic circle for signal distribution. ii) The *local Hybrid Zone* combines Wi-Fi and acoustic velocity components, which are projections of the actual speed, and applies a systematic theoretical method for accurate speed synthesis. This dual-stage design addresses the limitations of acoustic signals, such as narrow angular constraints and proximity requirements, while also mitigating spatial dependency issues inherent in Wi-Fi sensing. Implementing the Hybrid Zone significantly improves the accuracy and reliability of velocity, distance, and angle measurements, thereby enhancing object tracking across diverse scenarios.

IV. System Designs

A. System Overview

The system illustrated in Fig. 4 is composed of three main components: acoustic feature extraction, Wi-Fi feature extraction, and feature fusion.

1) Acoustic Feature Extraction: To effectively sense rapid motions over long distances, our system employs a triangular chirp signal, carefully designed to disentangle the effects of speed and distance on the acoustic signal. This approach not only ensures robust signal propagation but also enhances the resolution of motion sensing. We introduce a comprehensive theoretical model that leverages the chirp signal's properties in both the frequency and time domains, enabling precise tuning for optimal performance. This model serves as the foundation for accurately analyzing rapid movement dynamics, even in challenging scenarios with significant noise or interference.

In addition, the system incorporates a sophisticated microphone array, which plays a pivotal role in estimating the target's angle. This angular data, combined with distance measurements, allows us to derive an approximate global position for the target. This coarse-grained positional information is seamlessly integrated into the feature fusion module, providing a critical basis for advanced velocity synthesis and refined object tracking. By combining these elements, the acoustic feature extraction process delivers high accuracy and reliability for motion analysis.

2) Wi-Fi Feature Extraction: Our Wi-Fi feature extraction pipeline begins with calculating the CSI (Channel State Information) ratio, a key step for mitigating phase offsets and amplifying the dynamic components of the Wi-Fi signal. This ensures that the extracted features are more resilient to environmental noise and variations. Next, we apply the Short-Time Fourier Transform (STFT) to detect Doppler frequency shifts. These frequency shifts are converted into the Path Length Change Rate (PLCR), which directly quantifies the dynamic trajectory of the moving object. The PLCR is instrumental in capturing subtle motion variations over time, enhancing the precision of trajectory analysis.

To further refine our approach, we incorporate the Fresnel zone concept. By analyzing the interaction of the Wi-Fi signal with multiple ellipsoids, this method captures complex signal distortions caused by object movements. This integration of Fresnel zone analysis significantly improves our system's ability to interpret and track dynamic motion, making it effective for scenarios requiring fine-grained motion recognition.

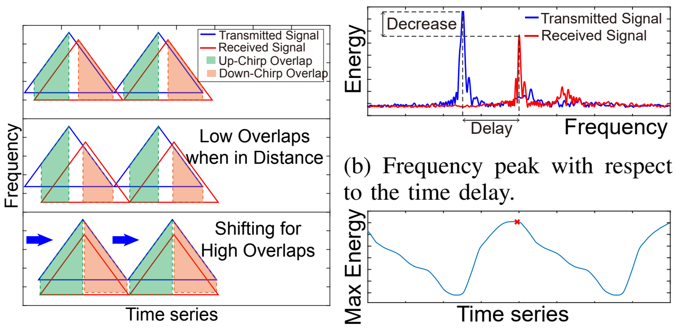
3) Feature Fusion: In the final stage, our system integrates data from both acoustic and Wi-Fi feature extractions, combining the acoustic circular zone and (Wi-Fi) Fresnel zone into the 'Hybrid Zone'. This process is divided into *global* and *local* perspectives. The global perspective considers overall patterns in the feature space, while the local perspective focuses on signal characteristics, ensuring comprehensive and accurate gesture recognition.

From a global perspective, we use angle-based and distance-based information purely from the acoustic system to estimate the object's coarse position. On the local scale, we merge velocity components identified by both Wi-Fi and acoustic signals. Each velocity component is projected according to its respective normal vector, and their combination provides an accurate measure of the user's speed through synthesis. This precise speed measurement is essential for describing dynamic movement changes across various timestamps during motion.

These global and local features are then fused into a time series format, forming the basis for our custom classifier. This classifier is meticulously crafted to recognize and distinguish between various gestures. By leveraging both global and local data, it serves as the culmination of our system, transforming complex data into actionable insights for gesture recognition. This approach ensures highly precise and reliable performance in real-world applications. Additionally, the classifier is designed to adapt to different environments, making it both versatile and robust for diverse usage scenarios.

B. Acoustic Feature Extraction

In this stage, we design a series of methods to realize acoustic feature extraction, and answer the following three



(a) Change of effective overlap with distance.

(b) Frequency peak with respect to the time delay.

(c) Find the time delay with maximum energy.

Fig. 6: Signal enhancement process.

questions: i) How to remove the interference of background noise? ii) How to realize long-distance sensing? iii) How to accurately determine the distance and speed? Next, we begin with our designed acoustic signals and then illustrate how to address the above-mention problems.

1) Sensing signals transmission: According to **Insight 1**, to distinguish the frequency offset caused by distance and speed, the crux is to design chirps with respect to different f, T, and B. In this paper, we utilize the triangular acoustic signals as shown in Fig. 5(a), which consists of up-chirps, down-chirps, and constant-frequency signals. To illustrate why our proposed chirp is valid, we take Fig. 5(a) as an example. Specifically, the effect of distance and speed on the frequency difference is caused by translation in the horizontal and vertical directions.

These two translations have different effects on the up-chirp, down-chirp, and constant-frequency signals. For the up-chirp, the resulting frequency shift is $f_d + f_v$, where f_d represents the frequency shift caused by the target's distance, and f_v represents the Doppler shift. The reason for this is that during the up-chirp, the instantaneous frequency is increasing, and the Doppler effect further shifts this increasing frequency upward. Therefore, both the distance-related shift (f_d) and the Doppler shift (f_v) contribute additively to the overall frequency shift. For the down-chirp, the resulting frequency shift is $f_d - f_v$, since the instantaneous frequency is decreasing, and the Doppler effect shifts the frequency downward as well. This results in the Doppler shift subtracting from the distance-related shift, which explains the negative sign in $f_d - f_v$. For the constant-frequency signals, the resulting frequency shift is f_v, because there is no frequency variation related to distance (i.e., no f_d component), leaving only the Doppler shift to affect the signal. By analyzing the difference between the up-chirp and down-chirp frequency shifts, we observe that the Doppler shift f_v cancels out, and the remaining frequency shift is $2f_d$, which is solely related to the target's distance. Hence, this enables us to distinguish between f_d and f_v effectively. A more detailed mathematical analysis of this result is provided later in the paper, where we derive the relationship between the frequency shifts and the target's distance and velocity.

Mathematically, the transmitter signal can be represented by

$$s_t(t) = \begin{cases} cos(2\pi f_0 t) + \cos\left(2\pi(f_0 t + \dfrac{Bt^2}{2T})\right), t \in \mathbb{S}_1 \\ cos(2\pi f_0 t) + \cos\left(2\pi((f_0 + B)t - \dfrac{Bt^2}{2T})\right), t \in \mathbb{S}_2 \end{cases} \tag{5}$$

where $\mathbb{S}_1 = [2nT, 2nT + T]$ and $\mathbb{S}_2 = [2nT + T, 2(n+1)T]$ respectively represent the first half and the second half of the periodic signal, and n denotes the n-th periodic signals. Next, we propose our method to extract acoustic features.

2) Background denoise: Aiming to address the first problem, we design the background denoise method. The fundamental principle of acoustic sensing is to analyze the feature of the signal reflected by the target. However, direct signals and signals reflected by other objects often act as interference, complicating the analysis and reducing accuracy. As illustrated in Fig. 5(b), multiple frequency peaks are respectively corresponding to the direct and dynamic signals. It is evident that the power of the direct signal is stronger compared to the reflected signal, as the latter undergoes greater energy attenuation. To mitigate the effects of these path deviations, we establish a ground truth by collecting acoustic signals with the object at a fixed, stationary distance. This involves removing background signals from the initial data for each chirp size, effectively isolating the target signal. After we perform the background denoise, the resulting spectrogram is shown in Fig. 5(c), where the direct signal is significantly suppressed, and we can obtain accurate frequency shifting with respect to the dynamic signal.

3) Signal enhancement: To address the second question, we design a signal enhancement method. In practice, acoustic signals are attenuated rapidly as they travel through the air, making it challenging to obtain accurate features over long distances. This problem becomes more serious for our hybrid chirp signals. Specifically, we derive location-related features by analyzing the frequency shift between the transmitted and received signals. However, as the distance between the smart speaker and the object increases, the signal delay grows, reducing the overlap between transmitted and received signals, as illustrated in Fig. 6(a). Accordingly, the frequency peak with respect to the object will decrease in Fig. 6(b). To counter this, we actively add a delay to the source signal, increasing the effective overlap area and enhancing the received signal strength. To find the optimal overlap area, we can iterate over the different time delays t_d and record the corresponding peak value $P(t_d)$. Fig. 6(c) illustrates the relationship between t_d and $P(t_d)$, where we can observe significant differences for different delays. Intuitively, a larger effective overlap area leads to higher $P(t_d)$ and more accurate features, and we have:

$$\hat{t}_d = \arg\max_{t_d} P(t_d).$$

In addition, when we involve such a delay, it is equivalent to translating the signal horizontally. Since f_d is also reflected as a translation of the signal in the horizontal direction, its estimation will be affected. To recover the feature, we record such extra delay $\hat{t}_d$ and then compensate the corresponding offset to the resulting distance, which is calculated by $\hat{t}_d c_s$.

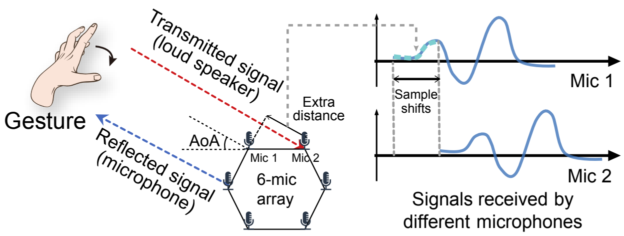


Fig. 7: The principle of AoA estimation.

4) Feature extraction: After enhancing the reflected signal, we analyze the time-frequency translation of the triangular signal to resolve location-related features. As depicted in Fig. 5(a), the equations for the periodic signals are given by:

$$\begin{cases} f_{up}(n) = f_d(n) + f_v(n), \\ f_{down}(n) = f_d(n) - f_v(n), \\ f_{single}(n) = f_v(n), \end{cases} \tag{6}$$

where n denotes the n-th periodic signal, and $f_d(n)$ and $f_v(n)$ are defined as

$$\begin{cases} f_d(n) = d(n)B/(Tc_s), \\ f_v(n) = v_s(n)f/c_s. \end{cases} \tag{7}$$

We provide a detailed mathematical derivation to show how the separation of Doppler shift and distance-related frequency shifts is achieved. By analyzing both up-chirp and down-chirp signals, we are able to isolate the Doppler shift $f_v(n)$ and the distance-related shift $f_d(n)$. To determine $d(n)$ (distance) and $v_s(n)$ (velocity from the acoustic perspective), we analyze $f_{up}(n)$, $f_{down}(n)$, and $f_{single}(n)$. In the time domain, we consider the integral relationship $d(n) = d(n-1) + 2v_s(n-1)T$ to further refine our calculations. We expand upon this by incorporating environmental variations in our analysis, ensuring that our method is robust under different conditions.

For acoustic velocity estimation, we combine time-frequency domain data using this equation:

$$v = \sqrt{v_1 \cdot v_3} + \min(e^{\alpha|v_2 - \sqrt{v_1 \cdot v_3}|}, 1)(v_2 - \sqrt{v_1 \cdot v_3}), \tag{8}$$

where α is a positive constant, and the components v_1, v_2, and v_3 are defined as

$$\begin{cases} v_1 = \dfrac{(f_{up} - f_{down})c_s}{2f}, \\ v_2 = \dfrac{\mathrm{d}}{\mathrm{d}t}\left(\dfrac{(f_{up} - f_{down}) \cdot c_s T}{2B}\right), \\ v_3 = \dfrac{f_{single} c_s}{2f}. \end{cases} \tag{9}$$

Equation (8) integrates these three velocity components, considering factors such as frequency shifts and time differentials, providing a comprehensive method for estimating the target's velocity, even under varying environmental conditions. This derivation substantiates our claim regarding the signal's effectiveness in distinguishing between Doppler shifts and distance-related shifts.

5) AoA estimation: Determining the global position of a target requires both its distance from a reference point and its angular orientation (AoA). We propose a method to calculate the AoA using a microphone array and the MUSIC algorithm [4], [27], inspired by the approach described in [7]. As shown in Fig. 7, when the target is sufficiently far relative to the spacing of microphones, the received signals can be approximated as parallel. This assumption allows us to calculate time delays between signals at different microphones, which correspond to specific arrival angles. To focus the AoA computation on target-reflected signals, we first eliminate the direct signal by eliminating the background noise. The AoA is then determined by detecting time delays, which are converted into phase differences across the microphones in the array.

To enhance precision, we utilize a hexagonal microphone array, which provides spatial diversity for accurate AoA determination. The phase difference between two microphones, based on their known positions and the speed of sound, can be expressed as:

$$\theta = \sin^{-1}\left(\frac{\Delta\phi_{ij} c}{2\pi f d_{ij}}\right),$$

where $\Delta\phi_{ij}$ and d_{ij} are the phase difference and distance between microphones i and j, c is the speed of sound, and f is the signal frequency. By combining precise signal alignment with the geometric properties of the hexagonal microphone array, we achieve a robust and accurate method for estimating AoA, enabling reliable global localization.

C. Wi-Fi Feature Extraction

In this phase, we extract Wi-Fi features from raw Channel CSI readings through physical processing. The CSI signal encapsulates various path elements, including static Line-of-Sight (LoS), environmental reflections, and dynamic paths from objects. The CSI at frequency f and time t is expressed as:

$$H(f,t) = H_s(f,t) + A(f,t)e^{-j2\pi\frac{d(t)}{\lambda}},$$

where $H_s(f,t)$ is the static component, and $A(f,t)$, λ, and $d(t)$ represent the attenuation, wavelength, and dynamic path length, respectively. To quantify motion, movement-induced path length changes are extracted by identifying the Doppler frequency shift (DFS) using the Short-Time Fourier Transform (STFT). Movement changes the path length of reflected signals, inducing a DFS. To determine DFS, we first compute the CSI ratio, then apply the STFT. We identify the DFS by extracting the highest energy frequency at each moment from the STFT spectrum, denoted as $f_{doppler}(t)$. The Fresnel model indicates that a full cycle in CSI magnitude equates to one wavelength change in path length. Hence, the PLCR of reflected signals is $v_w(t) = f_{doppler}(t)\lambda$. However, v_w doesn't exactly represent human velocity. Instead, it reflects the velocity projection along the Fresnel zone's normal line, requiring further calculations for true velocity determination. With pre-known transmitter ($\vec{\boldsymbol{l}}_t = (x_t, y_t)^T$) and receiver ($\vec{\boldsymbol{l}}_r = (x_r, y_r)^T$) positions, and the target's current position ($\vec{\boldsymbol{l}}_h = (x_h, y_h)^T$) and velocity ($\vec{\boldsymbol{v}} = (v_x, v_y)^T$), we can define PLCR v_w.

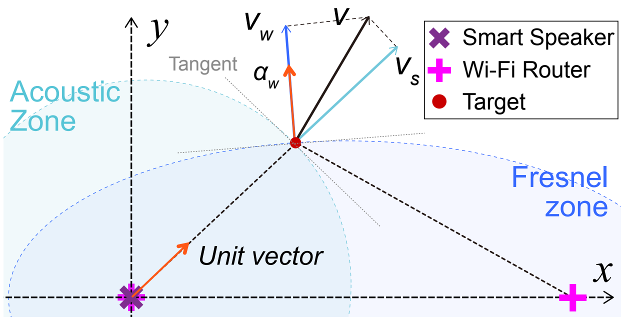


Fig. 8: Schematic representation of feature fusion.

D. Feature Fusion

The purpose of this stage is to fuse the acoustic and Wi-Fi features based on Hybrid Zone. In particular, the acoustic feature extraction stage provides $\{\theta, d, v_s\}$, while the Wi-Fi feature extraction stage provides $\{v_w\}$. To fuse the feature, we first use the interpolation method to synchronize acoustic and Wi-Fi features, and then propose how to use the hybrid area to fuse multi-modal features. Finally, we draw the trajectory of the target and realize gesture recognition.

1) Synchronization: To effectively analyze the current state using both Wi-Fi and acoustic signals, we need to ensure two crucial elements. First, the data from both signals must be synchronized, meaning that they should start and end concurrently. Second, since these two types of signals are recorded at different rates, it's necessary to manipulate the data so that they share the same time intervals. To accomplish this, we standardized both the Wi-Fi and the acoustic signals at a common sampling rate of 100 samples per second. The interpolation was then used to achieve uniform values. This process resulted in synchronized data with matching time intervals, thereby for further processing and analysis.

2) Hybrid Zone: In the global stage, we use the smart speaker as the origin and the direction from the smart speaker to the WiFi router as the x-axis to establish a coordinate system for simplicity.

Assume the target is at (x, y). We can transform the x and y components into d and θ components with $x = d\cos\theta$ and $y = d\sin\theta$. First, we focus on the local perspective of the Hybrid Zone. As Fig. 8 shows, v_s is the velocity observed by the smart speaker, and v_w is the velocity observed by both the smart speaker and the Wi-Fi router. We have two unit vectors, $\vec{\alpha}_s$ and $\vec{\alpha}_w$, representing the normal vectors of the target's current position from the Acoustic Circular Zone and Fresnel Zone, respectively. When the target moves at a *true* velocity v at the position (x, y) (*i.e.*, (d, θ)), the velocity observed by the smart speaker is a projection in the direction of the unit vector $\vec{\alpha}_s$ and can be written as

$$v_s = \alpha_x v_x + \alpha_y v_y, \tag{10}$$

where

$$\alpha_x = \frac{x_h}{\|\vec{\alpha}_s\|}, \ \alpha_y = \frac{y_h}{\|\vec{\alpha}_s\|}. \tag{11}$$

Recall that we have

$$\begin{cases} v_s = \alpha_x^{(s)} v_x + \alpha_y^{(s)} v_y, \\ v_w = \alpha_x^{(w)} v_x + \alpha_y^{(w)} v_y. \end{cases} \tag{12}$$

Then we can obtain the following equation:

$$\vec{v}_{obs} = \mathcal{L} \cdot \vec{v}, \tag{13}$$

where $\vec{v}_{obs} = [v_s, v_w]^T$, and $\mathcal{L} = \begin{bmatrix} \alpha_x & \alpha_y \\ \beta_x & \beta_y \end{bmatrix}$. In order to derive the user's velocity, we can solve $\vec{v}$ in Eq. (13) by

$$\vec{v} = (\mathcal{L}^T \mathcal{L})^{-1} \mathcal{L}^T \vec{v}_{obs}. \tag{14}$$

This allows us to calculate the target's true velocity $\vec{v}$ from the observed values, and forms the foundation for deriving accurate velocity estimations through Hybrid Zone.

V. Evaluation

This section delineates our experimental settings, spanning hardware and software considerations, and subsequently discusses the performance assessment of our system under varying conditions.

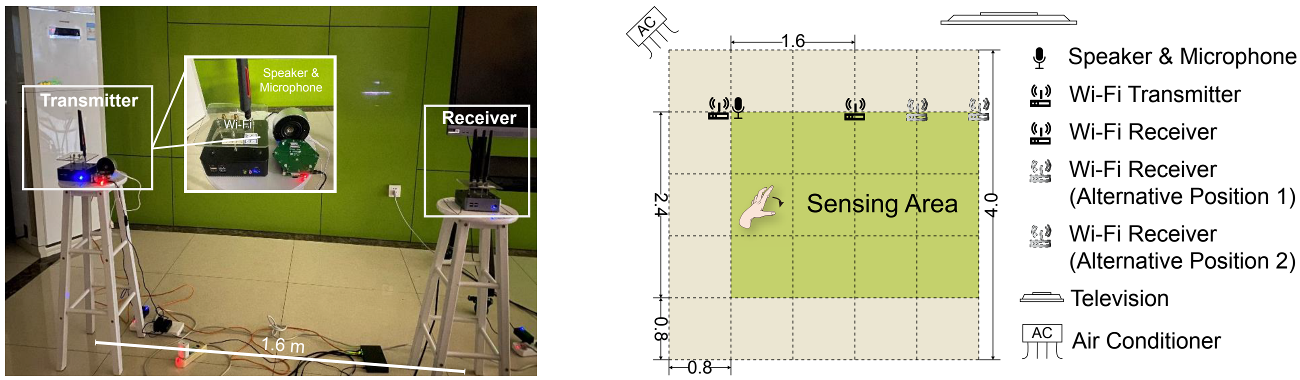
A. System Implementation

Our hardware setup, as shown in Fig. 9, encompasses various components necessary for our experiments. The Wi-Fi component of our system uses two devices equipped with Intel 5300 NICs: a single-antenna transmitter and a three-antenna receiver. For acoustic sensing, we employ a Seeed Respeaker circular array with six microphones, comparable to commercial smart speakers like Google Home and Amazon Echo. The microphone array, designed in a hexagon shape with a side length of 4.75cm and a sampling rate of 48kHz, is co-located with the Wi-Fi transmitter to support both acoustic and Wi-Fi functionalities. Both the microphone array and Wi-Fi transmitter are connected to a laptop for data collection, with the receiver positioned at a distance to enable effective sensing. Our system is implemented on a laptop equipped with an Intel i7-11800H CPU and 16GB of RAM. We utilize an SSH tool, which is compatible with the linux-802.11n protocol on Ubuntu 14.04.3, to manage packet transmission. The code is developed and executed within Matlab.

B. Overall System Performance

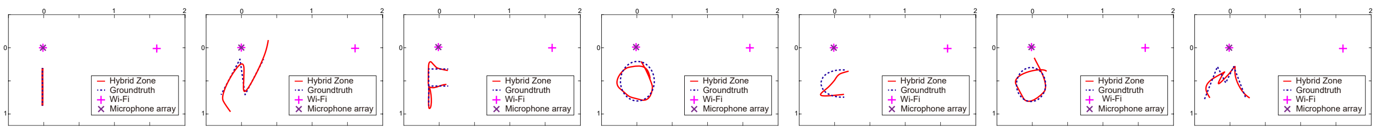
Assessing the Entire System Performance. To evaluate our proposed method's efficacy, we conducted an experiment where the term "INFOCOM" was sketched. Each letter of "INFOCOM" is represented in a series of figures (Figs. 10-12), with each figure illustrating a different letter. The gestures for 'I' to 'M' are showcased in three scenarios: with and without implementing the Hybrid Zone. The results are compared to a method relying solely on acoustic and Wi-Fi sensing, emphasizing the Hybrid Zone's effectiveness in overcoming the limitations of each individual technique.

Evaluation of Gesture Recognition Performance In addition to letter-based gestures, we define six new gestures for a more comprehensive evaluation: 'push', 'pull', 'leftward', 'rightward', O-shape, and V-shape. For each gesture, we collect



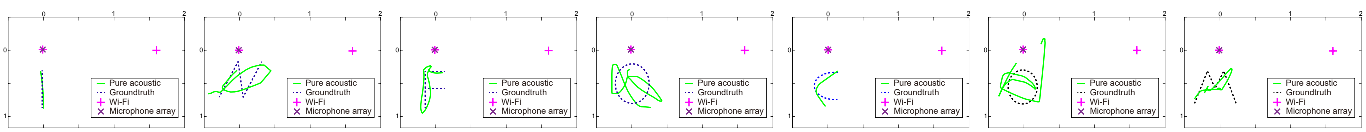
(a) Real-world measurement setup for the LoS scenario. (b) Schematic illustrating the LoS configuration.

Fig. 9: Experimental LoS scenario, showing (a) the real-world setup and (b) a schematic layout illustrating the LoS configuration.



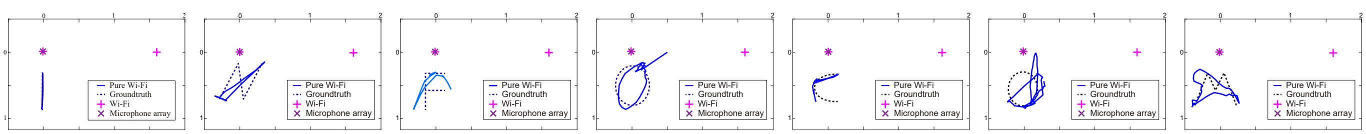
(a) Gesture 'I'. (b) Gesture 'N'. (c) Gesture 'F'. (d) Gesture 'O'. (e) Gesture 'C'. (f) Gesture 'O'. (g) Gesture 'M'.

Fig. 10: Tracking examples of different trajectories with Hybrid Zone.



(a) Gesture 'I'. (b) Gesture 'N'. (c) Gesture 'F'. (d) Gesture 'O'. (e) Gesture 'C'. (f) Gesture 'O'. (g) Gesture 'M'.

Fig. 11: Tracking examples of different trajectories with pure acoustic methods (SpeakerGesture [9]).



(a) Gesture 'I'. (b) Gesture 'N'. (c) Gesture 'F'. (d) Gesture 'O'. (e) Gesture 'C'. (f) Gesture 'O'. (g) Gesture 'M'.

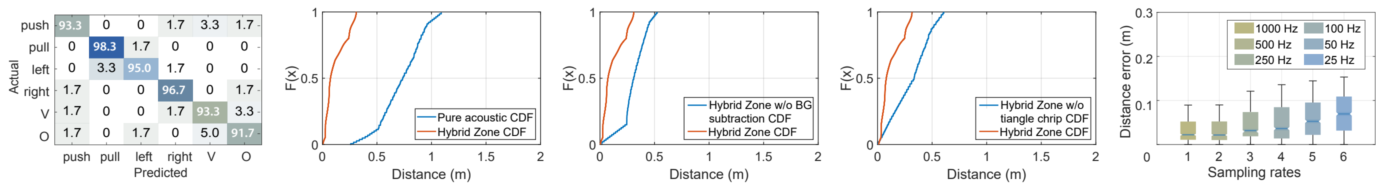
Fig. 12: Tracking examples of different trajectories with pure Wi-Fi method (Widar2.0 [28]).

60 samples to build a robust dataset. The goal is to utilize the Hybrid Zone approach to identify and analyze the unique signal patterns associated with these gestures. To demonstrate the effectiveness of Hybrid Zone, we evaluate its performance in two areas: gesture classification and motion tracking accuracy. Figures 13(a) and 13(b) depict these results separately, focusing on classification and tracking, respectively. Classification is achieved by training a machine-learning model to recognize gestures. This model leverages key features extracted from the data, enabling it to distinguish between gestures with high accuracy. The simulated data generated from the theoretical model enhances the robustness of the training process. With the theoretical model established in this paper, we generate simulated data to train a neural network for gesture recognition. Figure 13(a) shows classification results using a confusion matrix. The 'pull' gesture achieves the highest accuracy at 96.5%, while the O-shape gesture shows the lowest accuracy at 92%. All gestures maintain accuracy rates above 92%, highlighting the model's reliability. Figure 13(b) illustrates tracking precision through a Cumulative Distribution Function (CDF). Compared to a purely acoustic sensing approach, Hybrid Zone achieves 92% accuracy within a tracking error of 0.24m, significantly outperforming the acoustic method, which has an error of 0.78m.

TABLE II: Classification metrics with an additional column.

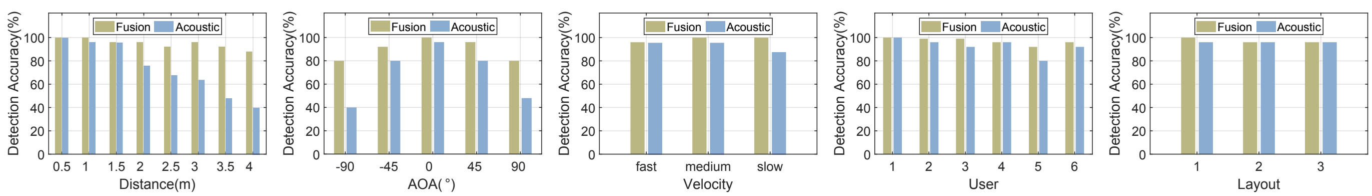
Class	Accuracy	Precision	F1-Score	Specificity	Confusion
push	0.933	0.965	0.949	0.951	0.067
pull	0.983	0.968	0.975	0.973	0.017
left	0.950	0.950	0.950	0.962	0.050
right	0.967	0.949	0.958	0.959	0.033
V	0.933	0.903	0.918	0.935	0.067
O	0.916	0.948	0.932	0.907	0.083

Detailed Analysis of Performance Metrics. Apart from evaluating overall accuracy, we use precision, F1-score, specificity, and confusion rate to obtain a more holistic picture of each



(a) Recognition accuracy. (b) Tracking accuracy. (c) BG subtraction. (d) Transmitted chirp. (e) Sampling rate.

Fig. 13: Overall performance.



(a) Impact of distance. (b) Impact of angle. (c) Impact of speed. (d) Impact of user. (e) Impact of layout.

Fig. 14: Impact of diverse parameters.

gesture class's performance. Accuracy measures how often the system correctly classifies gestures; precision gauges the system's ability to avoid false positives by comparing true positives to all predicted positives; and the F1-score, as the harmonic mean of precision and sensitivity, provides balance between correctness and coverage. Specificity is introduced to highlight how often negative instances are correctly rejected, aiding in recognizing the model's ability to distinguish one gesture from others. Finally, confusion rate identifies how frequently each gesture is mislabeled, revealing which motions are often mistaken for one another.

From the table, all six gestures maintain high accuracies, with the *pull* class reaching 0.983, alongside a specificity of 0.973. This combination underscores the model's effectiveness in both capturing true pulling gestures and minimizing incorrect classifications. Meanwhile, the *O* gesture has an accuracy of 0.916 and a slightly higher confusion rate of 0.083, suggesting partial overlap in motion characteristics with other classes. Nonetheless, its precision stands at 0.948, revealing that when the system does predict *O*, it is correct most of the time. These observations highlight an overall strong performance across gestures, while also indicating that targeted adjustments to the feature extraction process or classification architecture could further enhance the model's recognition consistency.

C. Factor Control Impact

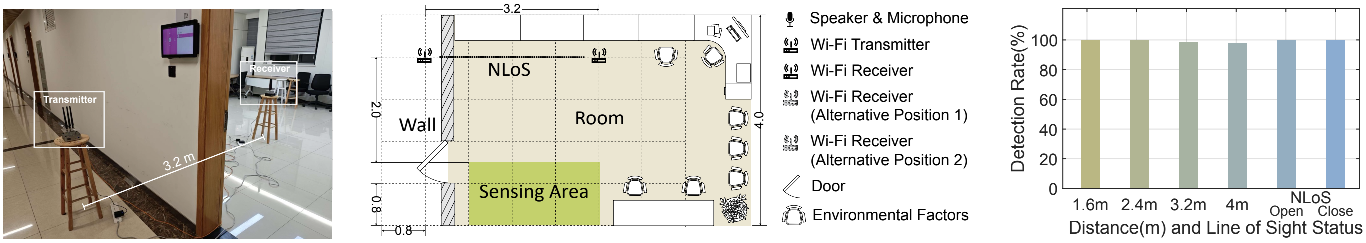
Impact of Background Subtraction. We examine background subtraction in enhancing our system's performance. As discussed earlier, most existing acoustic methods estimate the target position directly from the obtained frequency shift, neglecting any frequency shifts that aren't induced by distance. To address this, we incorporate a repetitive check after the coarse-grain distance estimation in our method. The results are shown in Fig. 13(c). The blue and orange lines follow similar trends, with the former appearing as a horizontal shift of the orange line. This supports our approach's effectiveness in reducing errors caused by incorrect distance estimations.

Impact of Chirps. To investigate how different chirp configurations affect our system's performance, we considered the use of triangle chirps. The reason we incorporate a down-chirp and a constant frequency signal, as mentioned earlier, is to better separate the factors of velocity and distance. Given that most existing acoustic-based research employs a linear up-chirp as the transmitted signal, we use the up-chirp portion of the triangle chirp for comparison. In other words, We focus solely on the first half of the triangle chirp, specifically the up-chirp, as depicted in Fig. 13(d), while omitting the second half. In this figure, the two lines closely follow each other, with the blue line slightly to the right of the orange one.

Impact of Wi-Fi Sampling Rate. We assess how different Wi-Fi sampling rates, ranging from 20Hz to 1000Hz, impact our system's performance. This evaluation is vital for gauging the system's adaptability in diverse Wi-Fi conditions. The outcomes of these assessments are presented in Fig. 13(e). The findings from these tests provide insights into the optimal sampling rates for achieving the best gesture recognition accuracy under various environmental conditions.

Impact of Distance. We conduct tests to investigate the variation in system performance with changes in the distance between the sensor and the subject. We consider eight different distances from 0.5 meters to 4 meters. For each distance, we ask a volunteer to repeat a gesture five times, with five repetitions per trial. As shown in Fig. 14(a), Hybrid Zone (blue bars) maintains an accuracy of 90% up to 3.5 meters. In contrast, pure acoustic sensing (red bars) fails to maintain its sensing capabilities (falling below 90% accuracy) after 1.5 meters. These results highlight the superiority of Hybrid Zone in capturing gestures over long distances.

Impact of Angle. We also test the effectiveness of Hybrid Zone under varying angles from -90 degrees to 90 degrees. Similar to the distance test, we ask the volunteer to repeat gestures 25 times at each angle. Unlike the acoustic-based methods that only achieve acceptable performance when facing the target directly, Hybrid Zone takes advantage of its velocity fusion capabilities, which integrate multiple sensing



(a) Real-world NLoS measurement environment.

(b) Schematic layout illustrating the NLoS configuration.

(c) Performance with transceiver separation and NLoS conditions.

Fig. 15: Experimental setup and evaluation results for the NLoS scenario and various transceiver distances.

modalities. This allows it to yield more precise distance and direction estimates and thus more accurate gesture predictions. The results in Fig. 14(b) show that Hybrid Zone maintains an accuracy higher than 80% at all angles.

Impact of Gesture Execution Speed. We examine the effect of the speed at which a gesture is executed on system performance. To achieve this, we asked the volunteer to perform the same gesture at three different speeds: slow, medium, and fast. As shown in Fig. 14(c), Hybrid Zone consistently maintained an accuracy above 95% across all speeds. However, purely acoustic methods underperformed at slow speeds. This drop in performance can be attributed to the slower execution, which results in negligible differences between consecutive timestamps. Consequently, it becomes challenging for the system to discern directional changes using pure acoustic sensing alone. This highlights the importance of considering gesture speed in the design and evaluation of gesture recognition systems.

Impact of User Variability. We conduct tests to measure the effect of user variability on our system's performance. We enlisted six volunteers to perform gestures 25 times each. The results, presented in Fig. 14(d), show how the performance of the system varied among users. All users achieved an accuracy higher than 90%, with User 1 having the highest accuracy at 100%, and User 5 having the lowest accuracy at 90%. Although acoustic signals also demonstrate good robustness across various users, the results may not be as reliable when users perform gestures differently, with an accuracy falling below 80% as shown in the figure.

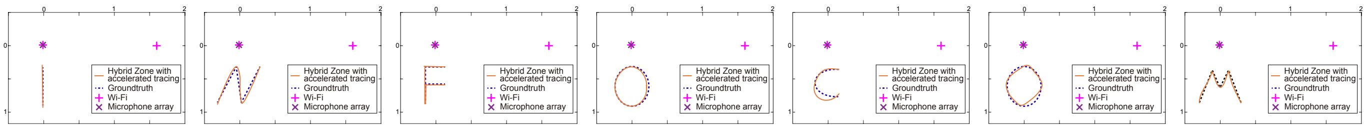
We evaluate the implications of sensor device placement on the effectiveness of our system, aiming to demonstrate that Hybrid Zone maintains its robustness across varying environments and scales. The initial experiment is conducted in three different layouts with environmental factors such as air conditioners and televisions, as shown in Fig.9. In these layouts, the distances between the Wi-Fi transmitter and receiver are set to 0.8, 1.6, and 2.5 meters. The results, presented in Fig.14(e), show that Hybrid Zone achieves an accuracy above 90% across all three layouts. Interestingly, we observed that the pure acoustic method remains unaffected by changes in layout. This is because the layout adjustments alter only the distance between Wi-Fi routers, while the placement of the acoustic sensing system remains constant, leaving its sensing accuracy largely unchanged. To further evaluate the scalability of Hybrid Zone, we conducted additional experiments with Wi-Fi transmitter-receiver distances of 0.8, 1.6, 2.4, 3.2, and 4 meters. Even at the largest distance of 4 meters, Hybrid Zone achieves a detection rate exceeding 97%, demonstrating its adaptability to larger scales. These results highlight the robustness of Hybrid Zone in varying spatial configurations and its potential for deployment in larger environments.

TABLE III: Comparison of time usage between pure method-based and accelerated tracing-based Hybrid Zone.

Data Points Number	10	500	10000	20000
Server (Data-driven, s)	3.86	193.12	3823.23	8292.94
RasPi (Data-driven, s)	5.17	242.81	4840.91	9683.40
Server (Method-based, s)	17.77	940.56	18417.59	36355.17
Gesture Drawing Time (s)	10.78	551.36	10925.14	21540.70

Impact of NLoS Conditions. This section examines whether the Hybrid Zone system maintains its performance under non-line-of-sight (NLoS) conditions. Specifically, we position the transmitter and receiver on opposite sides of a wall, eliminating direct line-of-sight between them. The user operates within the same room as the receiver, and we evaluate two NLoS scenarios: door open and door closed. When the door is open, the transmitted signal can reflect off the human body and reach the receiver via *reflected* propagation. In contrast, when the door is closed, both direct LoS and primary reflection paths are obstructed, and the signal reaches the receiver primarily through *diffraction* effects. This experiment is designed to evaluate the Hybrid Zone system's capability in handling both reflection-dominated and diffraction-dominated scenarios.

The experimental setup is depicted in Fig. 15, with Fig. 15(a) providing a real-world photo and Fig. 15(b) showing the corresponding schematic layout. The transmitter and receiver are separated by 3.2 meters, with the user performing gestures at a vertical distance of 2 meters from the devices. The room contains objects which may introduce multipath effects. The results, summarized in the last two columns of Fig. 15(c), show detection rates of 98% and 100% for the door-open and door-closed scenarios, respectively. These results provide insight into the robustness of the Hybrid Zone system under varying NLoS conditions. They demonstrate its strong capability to handle both reflection and diffraction propagation, ensuring reliable performance even in complex indoor environments. This robustness highlights the system's potential for practical deployment in real-world scenarios.



(a) Gesture 'I'. (b) Gesture 'N'. (c) Gesture 'F'. (d) Gesture 'O'. (e) Gesture 'C'. (f) Gesture 'O'. (g) Gesture 'M'.

Fig. 16: Tracking examples of different trajectories with accelerated tracing network.

D. AI Model on Embedded Devices Impact

In this section, we aim to determine if (a) an AI model can be used to improve tracking accuracy and (b) if it can be deployed on embedded devices. We begin by training a model to predict traces and classify gestures. Afterward, we deploy this model on an embedded device to reduce the response time.

1) Tracking Accuracy Improvement: We employ a basic Long Short-Term Memory (LSTM) model to learn and predict trajectories, consisting of a single LSTM layer and a dense layer. To create a comprehensive training set, we simulate 10,000 datasets, each representing a specific trajectory. For more accurately reflection of real-world gesture drawing, we account for the impact of acceleration and inertia on the generated trajectories. This ensures that the simulated data closely mimics the dynamics of actual gesture movements. After generating these realistic trajectories, we proceed to generate the data captured by predetermined WiFi routers and smart speakers. This data is then used for training the model, enabling it to effectively interpret and predict gesture-based trajectories in real-world scenarios. Again, we plot the seven letters of "INFOCOM" as shown in Fig. 16, where the orange trajectory illustrates the Hybrid Zone's tracing result using the network for prediction. We observe that Hybrid Zone achieves a more accurate result with this network.

2) Deployment on Embedded Devices: Given the large volume of acoustic and WiFi data in a home environment, uploading everything to a central server is impractical due to bandwidth limitations. Therefore, processing the data locally is preferable. However, most home environments lack high-speed processing equipment. To realize real-time recognition, computing tasks must be processed at the edge (*e.g.,* embedded devices). These devices are optimized for local execution, ensuring efficient and responsive data processing without extensive data transfers. This approach mitigates upload challenges and enhances performance by utilizing the computational capabilities of embedded devices.

We compare the time efficiency between the original Hybrid Zone and the AI-accelerated tracing deployed on embedded devices. We deploy the network introduced in the previous section to a Raspberry Pi (RasPi) 4B to test its time efficiency, which features four 1.5 GHz Cortex-A72 cores. We measured the time used to process the same datasets using method-driven and data-driven approaches and compared these times to the gesture drawing time. Time usage less than the drawing time is considered capable of achieving real-time performance. As shown in Tab. III, we conducted tests with datasets of 10, 500, 10,000, and 20,000 data points. These results demonstrate that deploying the model on embedded devices allows the Hybrid Zone to achieve real-time gesture recognition.

Overall, the experiments validate the feasibility of using embedded devices for efficient and real-time gesture recognition, highlighting the potential for practical applications in various real-world scenarios where low-latency performance is crucial.

VI. Discussion

Multi-user scenarios. The current implementation of Hybrid Zone faces challenges in multi-user environments due to its inability to distinguish between gestures from different users sharing the same space. This limitation can lead to ambiguities, particularly when overlapping gestures occur. Enhancing the system to address this issue is crucial for its applicability in shared or public spaces. Advanced learning techniques, such as user-specific signal separation or multi-instance learning, could help isolate individual gesture signals and improve recognition accuracy. Future work could explore integrating these techniques with existing Wi-Fi and acoustic signal processing capabilities to enable robust gesture recognition in multi-user contexts.

Data collection and adaptability. The reliance on manually collected data restricts Hybrid Zone's adaptability to new environments and scenarios. Collecting such data is not only time-consuming but also requires significant effort to ensure comprehensive coverage of all possible variations. Simulation-based training offers a promising alternative, as it can generate diverse datasets that capture a range of environmental and user-specific effects. By using simulations, the system could be trained to adapt to new environments without manual intervention, improving both efficiency and scalability. Future work will focus on simulation methods to generate realistic and varied training data that enhance system robustness.

Applications beyond gesture recognition. While Hybrid Zone is primarily designed for gesture recognition, its core fusion of Wi-Fi and acoustic signals has potential for broader applications in device-free sensing. For instance, this system could be applied to monitor macro activities, such as walking or running, as well as finer tasks like handwriting recognition or posture analysis. Additionally, with further refinement, it could enable applications in security, health monitoring, and smart home automation. To unlock this potential, the system's learning models and signal fusion algorithms need to be generalized to support diverse sensing tasks beyond gestures.

VII. Conclusion

This paper presented "Hybrid Zone", a novel system combining acoustic and Wi-Fi signals for gesture recognition. With a recognition accuracy of 94.69%, our system operates effectively using a single acoustic setup and a Wi-Fi transmitter-receiver pair, eliminating the need for prior knowledge of the target's initial position. Extensive experiments demonstrate its robust accuracy and unique capability to integrate the benefits of acoustic and Wi-Fi technologies. A key feature of Hybrid Zone is its adaptability, ensuring seamless deployment across diverse real-world applications, including both LoS and NLoS scenarios, while enabling real-time gesture recognition. Hybrid Zone proves flexible in environments ranging from smart homes, which demand low intrusiveness and high precision, to industrial contexts where resistance to environmental noise is essential. In conclusion, Hybrid Zone distinguishes itself through its high recognition accuracy, adaptability, and environmental versatility, representing a significant advancement in gesture recognition technology.

References

[1] Markets and Markets, "Gesture recognition and touchless sensing market by technology (touch-based, touchless), type, product (touchless biometric equipment, touchless sanitary equipment), industry and geography," Markets and Markets, Tech. Rep. BIZ5381, 2021.

[2] R. Gao, M. Zhang, J. Zhang, Y. Li, E. Yi, D. Wu, L. Wang, and D. Zhang, "Towards position-independent sensing for gesture recognition with wi-fi," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 2, pp. 1–28, 2021.

[3] X. Tong, Y. Wan, Q. Li, X. Tian, and X. Wang, "Csi fingerprinting localization with low human efforts," *IEEE/ACM Transactions on Networking*, vol. 29, no. 1, pp. 372–385, 2020.

[4] X. Tong, H. Li, X. Tian, and X. Wang, "Wi-fi localization enabling self-calibration," *IEEE/ACM Transactions on Networking*, vol. 29, no. 2, pp. 904–917, 2021.

[5] R. Gao, W. Li, Y. Xie, E. Yi, L. Wang, D. Wu, and D. Zhang, "Towards robust gesture recognition by characterizing the sensing quality of wifi signals," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 1, pp. 1–26, 2022.

[6] J. Liu, W. Li, T. Gu, R. Gao, B. Chen, F. Zhang, D. Wu, and D. Zhang, "Towards a dynamic fresnel zone model to wifi-based human activity recognition," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 7, no. 2, pp. 1–24, 2023.

[7] D. Li, J. Liu, S. I. Lee, and J. Xiong, "Fm-track: pushing the limits of contactless multi-target tracking using acoustic signals," *Proceedings of the 18th Conference on Embedded Networked Sensor Systems*, pp. 150–163, 2020.

[8] ——, "Lasense: Pushing the limits of fine-grained activity sensing using acoustic signals," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 1, pp. 1–27, 2022.

[9] ——, "Room-scale hand gesture recognition using smart speakers," in *Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems*, 2022, pp. 462–475.

[10] W. Mao, M. Wang, W. Sun, L. Qiu, S. Pradhan, and Y.-C. Chen, "Rnn-based room scale hand motion tracking," *The 25th Annual International Conference on Mobile Computing and Networking*, pp. 1–16, 2019.

[11] Q. Pu, S. Gupta, S. Gollakota, and S. Patel, "Whole-home gesture recognition using wireless signals," in *Proceedings of the 19th annual international conference on Mobile computing & networking*, 2013, pp. 27–38.

[12] Y. Zhang, Y. Zheng, K. Qian, G. Zhang, Y. Liu, C. Wu, and Z. Yang, "Widar3.0: Zero-effort cross-domain gesture recognition with wi-fi," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 8671–8688, 2021.

[13] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "Ccnet: Criss-cross attention for semantic segmentation," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 603–612.

[14] B. Zhang, M. Li, X. Xie, L. Fu, X. Tong, and X. Liu, "Rc6d: An rfid and cv fusion system for real-time 6d object pose estimation," in *IEEE INFOCOM 2022-IEEE Conference on Computer Communications*. IEEE, 2022, pp. 690–699.

[15] C.-F. R. Chen, Q. Fan, and R. Panda, "Crossvit: Cross-attention multi-scale vision transformer for image classification," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 357–366.

[16] Y. Tian, Y. Wang, R. Zheng, X. Liu, X. Tong, and K. Li, "Wstrack: A wi-fi and sound fusion system for device-free human tracking," in *Proc. of IEEE INFOCOM*. IEEE, 2023.

[17] R. Xiao, J. Liu, J. Han, and K. Ren, "Onefi: One-shot recognition for unseen gesture via cots wifi," in *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems*, 2021, pp. 206–219.

[18] X. Zhang, C. Tang, K. Yin, and Q. Ni, "Wifi-based cross-domain gesture recognition via modified prototypical networks," *IEEE Internet of Things Journal*, vol. 9, no. 11, pp. 8584–8596, 2021.

[19] B. Korany, C. R. Karanam, H. Cai, and Y. Mostofi, "Xmodal-id: Using wifi for through-wall person identification from candidate video footage," in *The 25th Annual International Conference on Mobile Computing and Networking*, 2019, pp. 1–15.

[20] X. Li, L. Chang, F. Song, J. Wang, X. Chen, Z. Tang, and Z. Wang, "Crossgr: Accurate and low-cost cross-target gesture recognition using wi-fi," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 1, pp. 1–23, 2021.

[21] A. Wang and S. Gollakota, "Millisonic: Pushing the limits of acoustic motion tracking," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 2019, pp. 1–11.

[22] W. Wang, J. Li, Y. He, and Y. Liu, "Symphony: localizing multiple acoustic sources with a single microphone array," in *Proceedings of the 18th Conference on Embedded Networked Sensor Systems*, 2020, pp. 82–94.

[23] M. Wang, W. Sun, and L. Qiu, "{MAVL}: Multiresolution analysis of voice localization," in *18th USENIX Symposium on Networked Systems Design and Implementation (NSDI 21)*, 2021, pp. 845–858.

[24] K. Qian, C. Wu, Z. Yang, Y. Liu, and K. Jamieson, "Widar: Decimeter-level passive tracking via velocity monitoring with commodity wi-fi," in *Proceedings of the 18th ACM International Symposium on Mobile Ad Hoc Networking and Computing*, 2017, pp. 1–10.

[25] Y. Zeng, D. Wu, R. Gao, T. Gu, and D. Zhang, "Fullbreathe: Full human respiration detection exploiting complementarity of csi phase and amplitude of wifi signals," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 2, no. 3, pp. 1–19, 2018.

[26] D. Wu, Y. Zeng, F. Zhang, and D. Zhang, "Wifi csi-based device-free sensing: from fresnel zone model to csi-ratio model," *CCF Transactions on Pervasive Computing and Interaction*, pp. 1–15, 2022.

[27] M. Kotaru, K. Joshi, D. Bharadia, and S. Katti, "Spotfi: Decimeter level localization using wifi," in *Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication*, 2015, pp. 269–282.

[28] K. Qian, C. Wu, Y. Zhang, G. Zhang, Z. Yang, and Y. Liu, "Widar2. 0: Passive human tracking with a single wi-fi link," in *Proceedings of the 16th annual international conference on mobile systems, applications, and services*, 2018, pp. 350–361.

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